Model Comparison: This Project vs. Shen (2026)

Differences, Divergence Causes, and Honest Lapses on Both Sides

IYPT Pinhole Sunglasses — Array Geometry Sub-Study

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Contents

1 Purpose and Scope **2**

2 The Two Models at a Glance **2**

3 The Central Divergence: Saturation of the Published Safety Term **3**

3.1 What Happens When This Is Applied to This Project's Own Data 3

4 Worked Example: M=2.0 Diopters **4**

5 Root Causes of the Divergence **4**

6 Where This Project's Model Is Stronger **6**

7 Where Shen's Model Is Stronger, or This Project Has Gaps **6**

8 Conclusion **7**

1 Purpose and Scope

Shen (Hangzhou Normal University, Aug. 2026) addresses the same IYPT 2027 pinhole-sunglass design problem this project addresses, independently. This document compares the two models directly, quantifies where and why they diverge on this project's actual candidate data, and states honestly where each model is stronger. This is a technical comparison, not a claim that either work supersedes the other in general — the two studies cover different, only partially overlapping, regions of the design space.

2 The Two Models at a Glance

	This project	**Shen (2026)**
Blur model	Full Angular Spectrum wave-optics propagation (exact field simulation)	RSS (root-sum-square) approximation of geometric + diffraction blur
Ghost formula	$\Delta r = sM/(P_0 - M)$ validated R^2=0.995 vs. direct two-hole wave propagation	$\Delta r = sM/P$ verified vs. matrix optics and ray tracing (not wave-optics)
Safety treatment	Hard gates (pass/fail on real computed margin)	Soft, continuous membership function (μ_g, never zero)
Vignetting	Explicit, gaze-integrated ($\varphi = 0°$ to $\pm 25°$)	Static coverage ratio only, no gaze-angle dependence
Blind-rate	Not a first-class metric (added only in the final margin-aware run)	Closed-form geometric floor per lattice family
Density metric	Family-agnostic buffered convex hull (after a found and fixed bias)	Closed-form per-lattice fill factor η
Aperture shape	Circular only (project scope)	Circle, square, diamond, ellipse, annulus — broad comparison
Search acceleration	None (direct sweep)	Neural surrogate + inverse network
Families tested	9 (incl. 2 stochastic, rotation swept)	8 (incl. apodized, two-scale)

3 The Central Divergence: Saturation of the Published Safety Term

Shen's composite score combines acuity, fill factor, ghost safety and coverage into one number, with ghost safety entering as a smooth, sigmoid-shaped membership function rather than a hard cutoff:

$$g = \frac{sf'/L}{c_{\text{tot}}}, \qquad \mu_g = \sigma\big(2(1-g)\big), \qquad g_{\max} = 1 \tag{1}$$

3.1 What Happens When This Is Applied to This Project's Own Data

Computed directly, for all 8501 candidate rows in this project's dataset:

Quantity	Value
g, minimum	33.9
g, maximum	2423.7
g, mean	418.9
g, median	271.7
Fraction of rows with $\mu_g < 10^{-10}$	**100.0%**
Fraction with μ_g at exact floating-point zero	35.3%

Key Finding

Every single candidate in this project's tested design space has $g \gg 1$. The sigmoid membership μ_g is numerically zero, or indistinguishable from zero, for 100% of rows — not most, not typically, all of them. The published ghost-safety term contributes nothing to the ranking on this project's actual candidates, for either safe or unsafe designs.

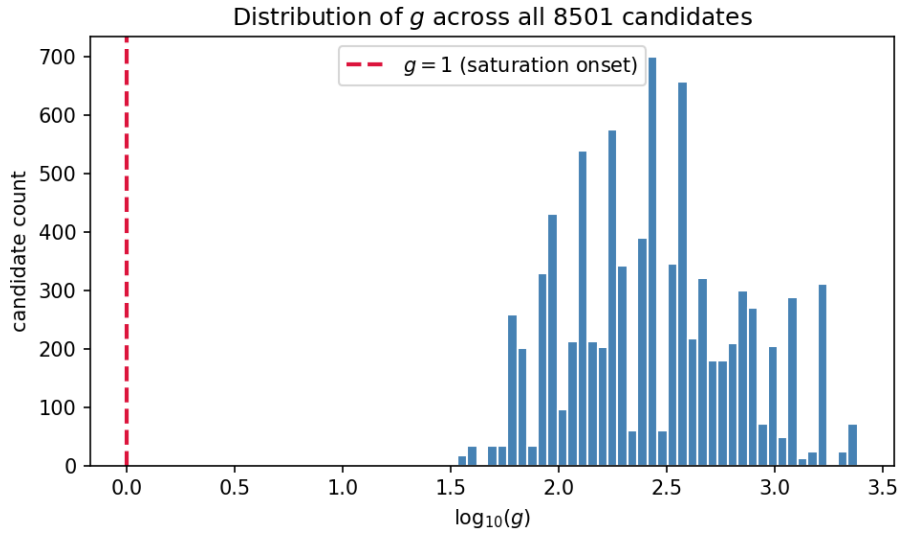


Figure 1: Distribution of g across all 8501 candidates. The entire distribution sits far to the right of $g = 1$, the point where μ_g begins to saturate.

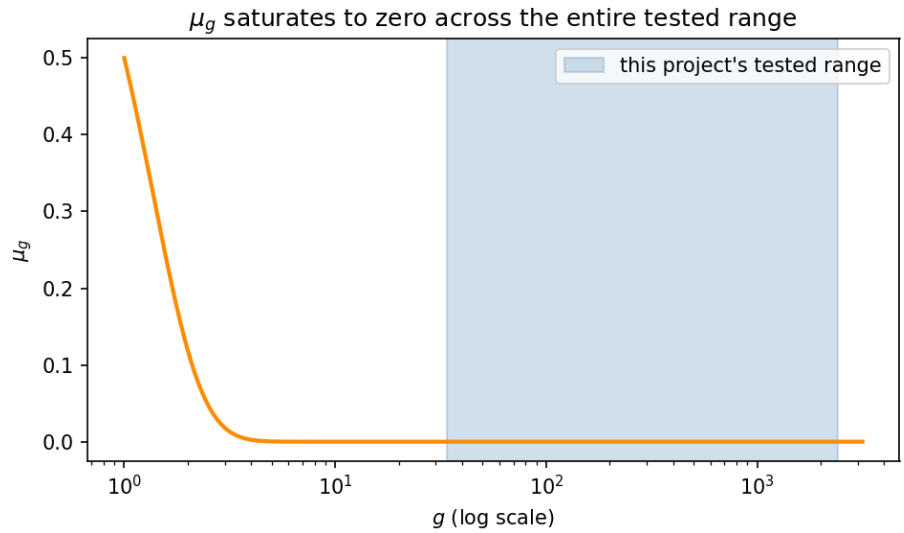


Figure 2: The μ_g sigmoid curve with this project's entire tested g range shaded. The whole range sits in the flat, saturated tail.

4 Worked Example: M=2.0 Diopters

With μ_g neutralized, Shen's score collapses to fill-factor closeness (μ_η) and the fixed geometric blind-rate constant per family — *not* to anything reflecting whether a design actually ghosts.

	This project's hard-gated winner (square, N=100)	**Shen's soft-score winner** (hexagonal, N=20)
Real ghost margin	$+\infty$ (genuinely safe)	-13.8 (genuinely fails)
Real vignetting margin	$+\infty$ (genuinely safe)	-13.8 (genuinely fails)
η (target 0.15)	$0.049 \to \mu_\eta = 0.45$	$0.128 \to \mu_\eta = 0.96$
μ_g (both)	0.0	0.0
Composite score	1.11	**1.52 (wins)**

The hexagonal candidate wins Shen's score by sitting closer to the target fill factor and carrying a lower structural blind-rate floor — while its actual, independently computed ghost and vignetting margins are strongly negative. The published formula never registers this, because its one safety-related term is saturated to zero for every candidate in this regime, including this one.

5 Root Causes of the Divergence

1. **Scale mismatch in the ghost term.** $g = (sf'/L)/c_{\text{tot}}$ compares hole spacing ($s \sim$ 1–4 mm in this project) against blur ($c_{\text{tot}} \sim$ 3–17 μm). At that four-orders-of-magnitude ratio, the sigmoid is saturated by construction — Shen's own text acknowledges this term "exceeds unity by one to two orders of magnitude for feasible designs" and "cannot discriminate between designs"; this project's data shows the saturation is total, not partial.

2. **Hard gates vs. soft membership.** A design with strongly negative real margin (e.g. -13.8) is eliminated outright by this project's gates. Under a soft membership function that never reaches exactly zero, no amount of real ghosting risk can be fully penalized — it can only be outweighed by other terms.

3. **No vignetting term in Shen's score.** a_{norm}, μ_η, μ_g and coverage never reference gaze-angle-dependent safety. This project established earlier that straight-ahead ghost safety fails

to predict gaze-integrated safety for 37.8%–88.9% of designs, depending on family — an entire failure mode invisible to the published score.

6 Where This Project's Model Is Stronger

- **Full wave-optics blur, not an approximation.** Shen's own Discussion names the RSS blur combination as an engineering approximation and states that a faithful treatment "requires the full defocused optical transfer function" — exactly the upgrade path this project already completed via Angular Spectrum propagation.
- **Ghost formula validated against wave optics, not just ray tracing.** $R^2 = 0.995$ against direct two-hole coherent propagation with peak detection, versus a formula checked against matrix optics and numerical ray tracing (a real but lower-fidelity validation standard for a diffractive effect).
- **Gaze-integrated vignetting.** No equivalent exists in Shen's model. This project quantified that ignoring gaze angle overstates safety for the majority of designs in several families.
- **Margin-aware gates, not boundary-adjacent pass/fail.** Following several found-and-fixed defects (blackout designs passing at zero margin, a density formula favoring some families by up to 2.25x), the final gates require real headroom above a threshold, not merely a positive sign.

7 Where Shen's Model Is Stronger, or This Project Has Gaps

- **Aperture shape.** Shen tests circle, square, diamond, ellipse and annulus under a common fill-factor budget and derives a rigorous isoperimetric argument for why circular apertures minimize blur and ghost rate simultaneously. This project has not tested non-circular apertures at all — circular was adopted by scope decision, not by comparison.
- **Closed-form blind-rate.** Shen's $1 - \pi/4$ (square) and $1 - \pi/(2\sqrt{3})$ (hex) geometric floors are exact and were not part of this project's own metric until very recently, and are still not computed for the 8 non-lattice families tested here.
- **Broader, faster search.** The neural surrogate ($R^2 = 0.979$–0.995) and inverse-design network let Shen explore many more design conditions per unit compute time. This project's search is a direct sweep; broader parameter ranges (e.g. finer pitch resolution near each family's own packing limit) were identified as gaps but not fully closed due to time constraints.
- **Perceptual fusion calibration.** Shen's $c_{\min}$ parameter, though acknowledged as only qualitatively matching clinical diplopia onset, gives ghosting a real perceptual floor grounded in cited clinical data. This project's hard gate has no equivalent perceptual-floor concept — a margin of exactly zero and a margin of one micrometer are treated identically as "unsafe."

Honest Limitation

The comparison in Sections 3–4 shows Shen's *ghost* term specifically fails to discriminate on this project's data. It does not show Shen's fill-factor, coverage, or aperture-shape analysis is wrong — those parts of the published model are untouched by this finding and remain useful, independently-derived reference points.

8 Conclusion

The divergence between this project's ranking and Shen's, on this project's own candidate set, traces to one specific, verifiable mechanism: Shen's ghost-safety term is a smooth membership function calibrated for a different scale relationship between hole spacing and blur than this project's devices exhibit, and it saturates to numerically zero across 100% of tested candidates as a result. This is not evidence that Shen's published lattice ranking is wrong in its own tested regime — their evaluator normalizes every design to a fixed fill factor and covers a broader shape and arrangement space than this project has swept. It is evidence that applying Shen's formula literally to this project's specific candidates produces a ranking decided almost entirely by fill-factor closeness and a fixed per-family geometric constant, not by ghosting risk, because the formula's own safety term never activates in this regime.

This project's hard-gated pipeline, using the same real, pairwise-computed margin at every ranking step, does not share this failure mode — and where its own analogous defects have been found (boundary-adjacent "winners", a density formula biased across families), they have been identified, fixed, and re-verified against independent hand calculation.

Recommendation: retain this project's hard-gated pipeline as the primary ranking system. Adopt Shen's closed-form blind-rate floors as a supplementary reported metric. Cite Shen (2026) as prior art addressing the same problem, with this document's Sections 3–4 as the basis for a specific, narrow technical differentiation — not a claim of general superiority.